An important practical aspect related to tokens is billing. Many LLM services charge users based on the number of tokens processed, which includes both the tokens in your input prompt and the tokens generated in the model's output. It's important to consider this for both managing and optimizing the cost of interacting with these models.

To better understand the concepts of tokenization, let's visualize the process using a simple prompt as our example. Figure 5.2 illustrates the journey of our prompt through various stages of tokenization and processing, together with an example prompt.

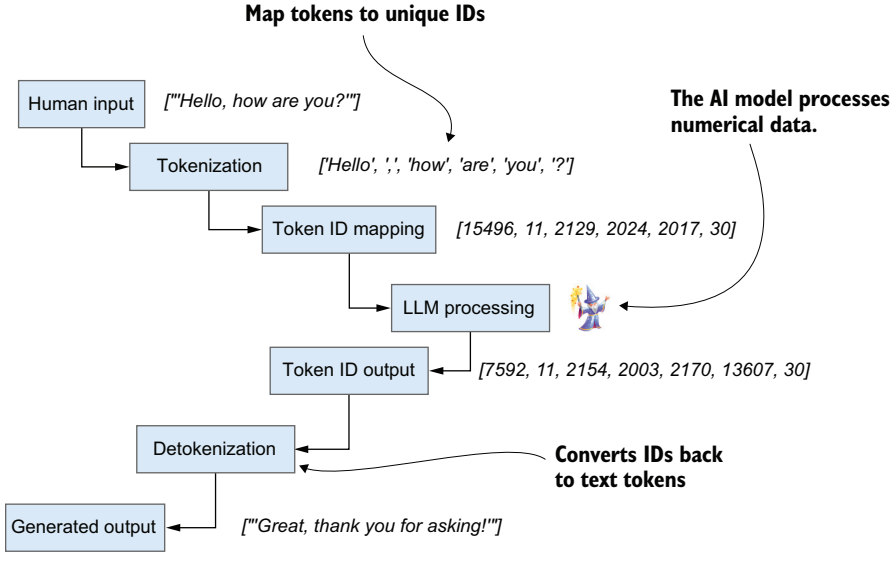


Figure 5.2 The end-to-end process of how human input is transformed, processed by an LLM and converted back into human-readable output

The code on which LLMs operate is a numerical representation in the form of floating-point numbers. They receive numbers as input and produce numbers as output, which are then translated back into human-readable formats like text or images. This numerical intermediary is the key to the LLM's ability to process and generate human-like responses.

The pipeline consists of seven key stages:

1. *Human input*—The raw text provided by the user via an input prompt.
2. *Tokenization*—The process of splitting the input text into tokens and converting them to numerical IDs. These numerical IDs come from a predefined vocabulary or tokenizer model associated with the LLM.
3. *Token IDs*—The numerical representation of the input, ready for LLM processing.
4. *LLM processing*—The core stage where the AI model analyzes the input and generates a response.